

AI-Powered Study Planner: A Smarter Way to Study

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Abstract :

Studying for exams can be overwhelming, especially when trying to figure out what to focus on and how to manage time effectively. This project introduces an AI-powered study planner that helps students create a smart study schedule. Using Natural Language Processing (NLP), the system extracts important topics from PDF lecture notes and assigns them a priority level. Based on the available time before an exam, it organizes a study plan using the Pomodoro technique (25-minute study sessions with breaks) and reserves the last two hours for revision with flashcards. This makes studying more structured, efficient, and stress-free.

Introduction :

Students often struggle with organizing their study time, not knowing which topics to prioritize or how to make the best use of their time before an exam. This project aims to solve that problem by using AI to generate a personalized study schedule. By analyzing the content of lecture notes, the system identifies key topics, ranks them by importance, and allocates study time accordingly. The goal is to remove the guesswork from studying and help students focus on what matters most.

NLP Models :

- **Gemini-1.5-Flash** - Extracting key topics from PDF lecture notes and ranking them by importance.
- **BERT** - Helps in understanding the context of extracted topics for better prioritization.
- **TF-IDF** - Assists in identifying the most relevant terms from lecture notes.
- **GPT-based models** - Utilized for generating structured study schedules based on the extracted topics.

Datasets :

<https://www.kaggle.com/datasets/saurabhprajapat/chatbot-training-dataset>

Input :

- Exam date and time (provided in YYYY-MM-DD HH:MM format).
- PDF lecture notes containing study material.

Expected Output :

2025-03-03 12:00 - Study Quantum Mechanics (25 min)
2025-03-03 12:25 - Break (5 min)
2025-03-03 12:30 - Study Newton's Laws of Motion (25 min)